

Coastal Hydrodynamics III

Course : CE60222

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Content

- Wave-induced Set-up and Currents
- Wave-induced Mass Flux or Momentum
- Radiation Stress
- Radiation Stress from Linear Wave Theory
- Radiation Stress from Breaking Wave

Wave-induced set-up and currents

Wave-induced mass flux or momentum:

- **Propagating waves** not only carry **energy across** the ocean surface, but **momentum as well**.
- **Momentum** is defined as the **product** of **mass** and **velocity**. It can be thought of as **mass in motion** or a **mass transport** or **flux**: a water particle has a **mass**, and if the **particle is moving**, it has **momentum**.
- **Momentum** per unit volume can thus be written as the **product** of the **mass density** ρ and the **velocity** $\vec{u} = (u_x, u_y, w)$ of the water particles.
- **Momentum** (per unit volume) $\rho\vec{u} = (\rho u_x, \rho u_y, \rho w)$ is a **vector quantity**, a quantity which is fully described by both **magnitude** and **direction**.
- The **direction** of the **momentum vector** is the same as the **direction** of the **velocity vector**.

Wave-induced mass flux or momentum

- This **momentum** is the **net flux of mass** between wave trough and **wave crest** associated with **wave propagation**.
- The **total amount of wave momentum per unit surface area** in the direction of wave propagation is obtained by **integration** over the **depth** and **averaged over time** (q) this gives (with u the horizontal orbital velocity in the wave propagation direction):

$$q = \overline{\int_{-h}^{\eta} \rho u dz}$$

- The **integral** above equation can be computed for **a single harmonic component (non-breaking)** by substituting the **velocity according to linear wave theory** at $z = 0$ and integrating from $z = 0$ to the instantaneous **surface elevation** $\eta = a \cos \omega t$.

- Then for the **mean momentum** in a **plane perpendicular** to the **wave propagation** direction per **unit surface area** ($q_{\text{non-breaking}}$) is given by,

$$q_{\text{non-breaking}} = \overline{\int_0^a \cos \omega t \rho \frac{a\omega}{\tanh kh} \cos \omega t dz} = \overline{a \cos \omega t \rho \frac{a\omega}{\tanh kh} \cos \omega t}$$

$$= \frac{\rho a^2 \omega}{2 \tanh kh} = \frac{\rho g a^2}{2c} = \frac{E}{c}$$

- **Above Equation** is **valid outside** the **surf zone**.
- **In the surf zone** the **mass flux** is substantially **larger** than **outside** the **surf zone**. (q_{drift})
- It is assumed to consist of **two parts**,
 - one due to the **progressive character** of the waves ($q_{\text{non-breaking}}$) and
 - the other due to the **surface roller** in **breaking** waves (q_{roller})

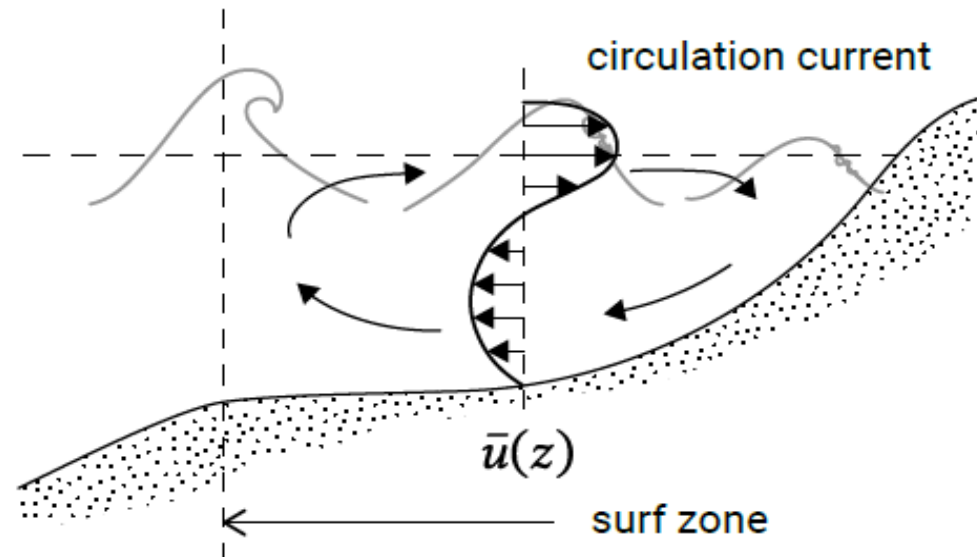
$$q_{drift} = q_{non-breaking} + q_{roller} = \frac{E}{c} + \frac{\alpha E_r}{c}$$

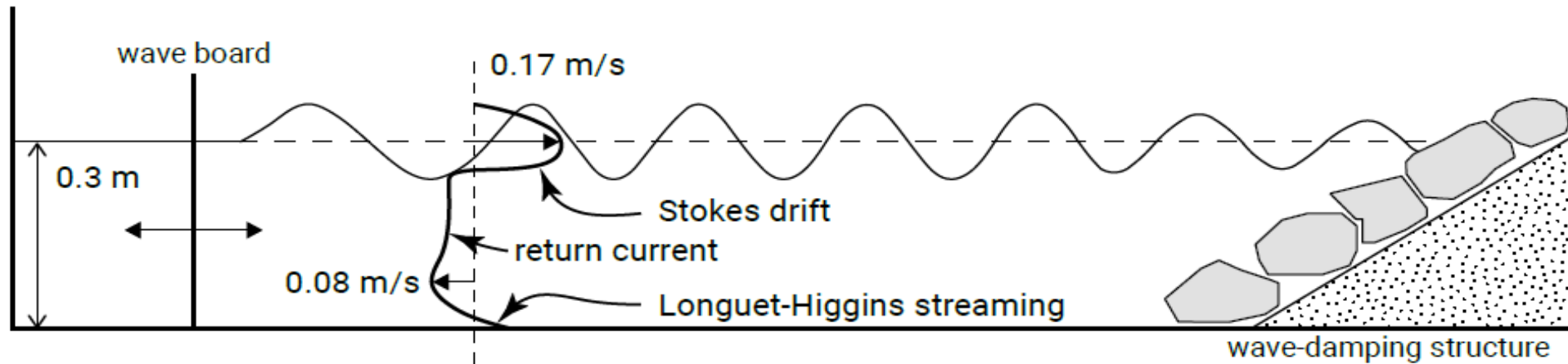
- In this equation, E_r is the **roller energy**.
- The **first part** of the **right-hand side** is the **mass flux** for **non-breaking** waves
- The **second part** accounts for the contribution of **the mass of the surface roller**, factor α to be in the range of **0.22 to 2**
- In the case of a **closed boundary**, like a **coastline**, there is a **zero net mass transport** through the **vertical**, as otherwise **water** would **increasingly pile up against the coast**.
- This means that there must be a **net velocity below the wave trough level** to **compensate** for the **flux above the wave trough level**: **a return current**.

- The **cross-shore depth-mean velocity below the wave trough** level must compensate for the **mass flux perpendicular to the shore** and is therefore given by:

$$U_{below\ trough} = -\frac{q_{drift,x}}{\rho h} = -\frac{q_{drift} \cos \theta}{\rho h}$$

- In **breaking waves**, the **mass transport** towards the coast between wave crest and wave trough may be quite large, resulting in rather large seaward-directed velocities under the wave trough level.
- The large **return current** in the surf zone is called **undertow**.





- Velocities averaged over a wave cycle under propagating waves in a wave flume. Even in this case of non-breaking waves, there is a **small return current**.
- Note that the term '**undertow**' is only used for the **larger return current** under breaking waves.
- Also in the **two-dimensional case** of a laboratory wave flume, the same mass of water has to return to the 'sea' again.
- In the lower part of the water column, this gives a **return flow**.
- The figure also shows **Longuet-Higgins streaming** close to the bed.

Radiation Stress

- **Radiation stress** is a measure of the **momentum transport** due to **wave action** in coastal hydrodynamics.
- It **arises** from the **changes in pressure and velocity fields** caused by waves and is represented by a **stress tensor** with both **normal and shear components**.
- **Radiation stress** is the name given to the **depth-integrated and wave-averaged flow (or flux)** of **momentum** due to waves.
- It was **first defined** by **Longuet-Higgins and Stewart (1964)** as the **excess momentum flux** due to the **presence of waves**.
- If there is **change in wave-induced momentum flux (radiation stress)** from **one location** to another, wave forces act on the fluid, impacting **mean water motion and levels**.

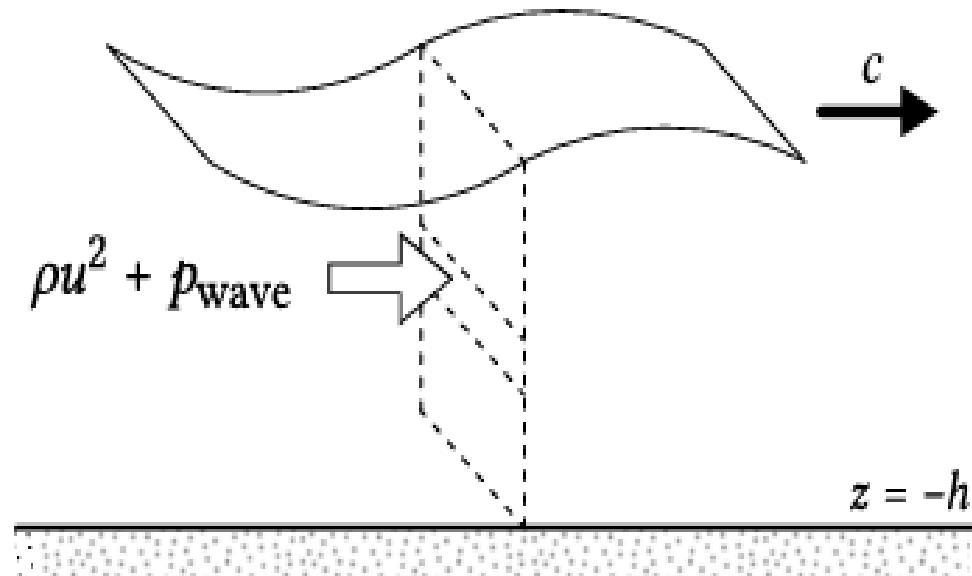
These wave forces are responsible for:

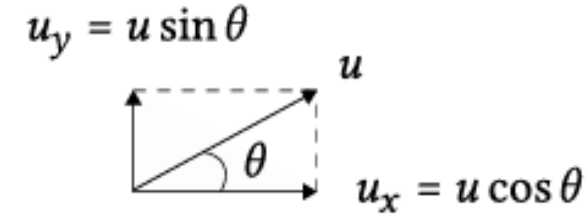
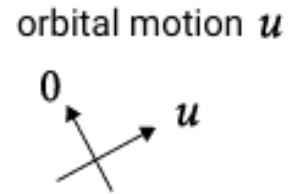
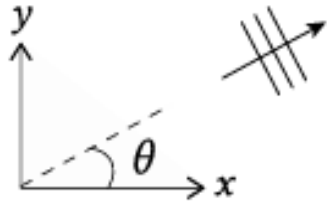
- **lowering** the **mean water level** in the **shoaling zone** (**set-down**).
- **raising** the **mean water level** in the **surf zone** (**set-up**).
- driving a **longshore current** in the case of **waves obliquely approaching** the **shore** (1 m/s under some conditions).

The wave-induced horizontal flux of momentum through a vertical plane at a given location consists of:

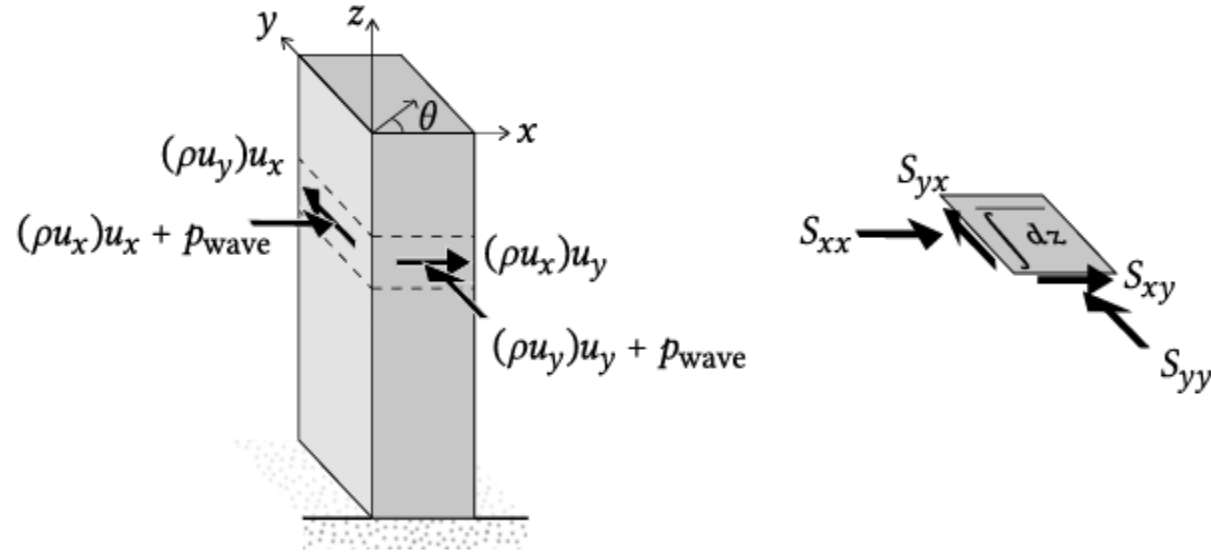
- the **transfer of momentum** $\rho \vec{u}$ **through** that **plane**, with the **particle velocity normal** to that **plane**;
- the **wave-induced pressure force** acting on the **plane** due to the **wave-induced pressure** p_{wave} in the water.

- First assume that the **plane under consideration** is **perpendicular** to the **wave propagation**.
- The **wave-induced pressure** p_{wave} acts by definition **normal** to the **plane**.
- Besides, at **every height above the bed**, the **particle velocity** u transports momentum ρu through the **plane** (per unit **crest length**).





- In **Figure** the **wave propagates at an angle** with the **x-axis**.
- The **particle velocity** has a component u_x in the **x-direction** and a component u_y in the **y-direction**.
- The **particle velocity** u_x transports both **x-momentum** ρu_x and **y-momentum** ρu_y



- The **transport of x -momentum** by the **particle velocity u_x** through a **vertical plane perpendicular to the x -axis** (per unit time and per unit area) is **$(\rho u_x)u_x$** and the transport of **y -momentum** through the **plane** is **$(\rho u_y)u_x$** .
- Figure shows the momentum transport and radiation stress components in x -direction and y -direction at a certain point in x,y -space.

- The **momentum transport** through the entire **plane per unit crest length** is obtained by **integration over the depth** from bottom to instantaneous water surface.
- **Time-averaging** yields, for the **total wave-averaged transport of x -momentum** in the **x -direction** or the **radiation stress S_{xx}** :

$$S_{xx} = \underbrace{\int_{-h_0}^{\eta} (\rho u_x) u_x dz}_{\text{part due to advection by horizontal particle velocity}} + \underbrace{\int_{-h_0}^{\eta} p_{\text{wave}} dz}_{\text{pressure part}}$$

- S_{xx} acts **normal** to the **considered plane** and is therefore a **normal component** of the **radiation stress**.
- It is **equivalent** to a **normal stress** acting in the **x -direction**.

- The **radiation stress** component S_{xy} (the **shear component** of the radiation stress) is defined as the **transport of x -momentum** in the **y -direction**.
- It acts as a **shear stress** on the **plane** (it is directed in the **x -direction**, and works on the plane **normal** to the **y -direction**) and is given by:

$$S_{xy} = \overline{\int_{-h_0}^{\eta} (\rho u_x) u_y + \tau_{xy} dz}$$

- The **shear stress** due to waves is **zero** for an **irrotational ideal fluid**, so that we have:

$$S_{xy} = \overline{\int_{-h_0}^{\eta} (\rho u_x) u_y dz}$$

- It consists of a **contribution** due to **advection** by the **horizontal orbital velocity** only.
- Note that for the **special case** of **normally incident waves**, the **x-direction** is the **wave propagation direction**, and u_y and hence S_{xy} are **zero**.
- Considering a **plane normal** to the **y-direction**, it is found for the **momentum fluxes** in the **y-direction**, the **normal component**:

$$S_{yy} = \underbrace{\int_{-h_0}^{\eta} (\rho u_y) u_y dz}_{\text{part due to advection by horizontal particle velocity}} + \underbrace{\int_{-h_0}^{\eta} p_{\text{wave}} dz}_{\text{pressure part}}$$

and the **shear component**:

$$S_{yx} = \overline{\int_{-h_0}^{\eta} (\rho u_y) u_x dz}$$

- In the **special case** that the **x-direction** is the **wave propagation direction** $u_y = 0$, S_{yy} reduces to the **pressure part** and $S_{yx} = S_{xy} = 0$.

Radiation Stress from linear wave theory

- Wave propagating in positive x -direction so that $u_x = u$ and $u_y = 0$.

$$S_{xx, \text{ horizontal particle velocity}} = \overline{\int_{-h_0}^{\bar{\eta}} (\rho u^2) dz} \approx \int_{-h_0}^0 \rho \bar{u}^2 dz = nE$$

where:

$S_{xx, \text{ hor. part. vel.}}$ = radiation stress in the wave propagation direction due to advection of momentum by the horizontal orbital motion (N/m)

n = ratio of group velocity and phase velocity

E = wave energy in the water column per m^2 J/ m^2

- In the case of **irregular waves**, this **equation** can be applied using $E = 1/8 \rho g H_{rms}^2$.
- For the **mass flux**, we found a **zero contribution** for every level **below the wave** trough because $\bar{u} = 0$.

- The **momentum flux**, however, is **non-zero** for the **entire water depth**, since $\overline{u^2} \neq 0$. It varies in principle over the depth.
- Only in **shallow water**, where the **horizontal orbital velocity** is **uniformly distributed**, the particle velocity part of the **radiation stress** is **uniformly distributed**.
- The **pressure** for **any level below the trough** would give a **zero time-averaged** result according to **linear theory**.
- There is, however, a contribution to the **time-averaged wave-induced pressure** $\overline{p_{wave}}$ due to the **vertical flux of momentum** by the **vertical fluid motion**.
- This may be thought of as the **vertical oscillatory fluid motion**, helping carry the **weight** of the **water column**, and is given by $-\overline{\rho w^2}$, in which w is the **vertical orbital velocity**.

- The **contribution** to the **radiation stress** can be found by substituting the **linear equation** for **w** and **integrating** to the **mean water level**. We find:

$$S_{xx, pressure, 1} = - \overline{\int_{-h_0}^{\bar{\eta}} \rho w^2 dz} \approx - \int_{-h_0}^0 \rho \overline{w^2} dz = (n - 1)E$$

- Furthermore, there is a **contribution** to the **radiation stress** due to the **pressure fluctuations** between **wave trough** and **wave crest level**.
- We assumed that the **pressure** between **wave trough** and **crest level** fluctuates as $\tilde{p} = \rho g \eta$.
- This gives the following **net contribution** to the **radiation stress**:

$$S_{xx, pressure, 2} = \overline{\int_0^{\bar{\eta}} \rho g \eta dz} = \frac{1}{2} \rho g \overline{\eta^2} = \frac{1}{2} E$$

- This gives for the **pressure part** of the **radiation stresses**:

$$S_{xx, pressure} = \underbrace{\frac{(n - 1)E}{\text{contribution due to (depth-varying) vertical momentum flux}}}_{\text{contribution due to (depth-varying) vertical momentum flux}} + \underbrace{\frac{1/2E}{\text{contribution at top of water column}}}_{\text{contribution at top of water column}} = (n - 1/2)E$$

- The term $\overline{w_2^2}$ **varies in principle** with the **water depth**, whereas the **other part** is located at the **top of the water column**.
- In **shallow water** ($n = 1$) the vertical orbital velocities are zero and the pressure part of the **radiation stresses** **reduces** to a **contribution** $S_{xx, pressure} = 1/2E$ at the **top of the water column**.
- In **deep water**, where $n = 1/2$, the **pressure component** to the **radiation stress** is $(n - 1/2)E$.

- The **total radiation stress** in the **wave propagation** direction (**x-direction**) is now:

$$S_{xx} = S_{xx, \text{pressure}} + S_{xx, \text{horizontal particle velocity}} = (n - 1/2)E + nE$$

- The **stresses normal** to the **wave propagation** direction S_{yy} consist of the **pressure part** $(n - 1/2)E$ only. The **shear stresses** are **zero**.

It may be concluded that:

- The **pressure part** of the **radiation stress** is equal to $(n - 1/2)E$. Since **pressure** is a **scalar**, this term is part of **radiation normal stresses** in all directions.
- The **magnitude** of the **advective part** of the **radiation stress**, the part due to **transport of momentum by the particle velocity**, is nE . It is by definition in the **wave propagation direction**.

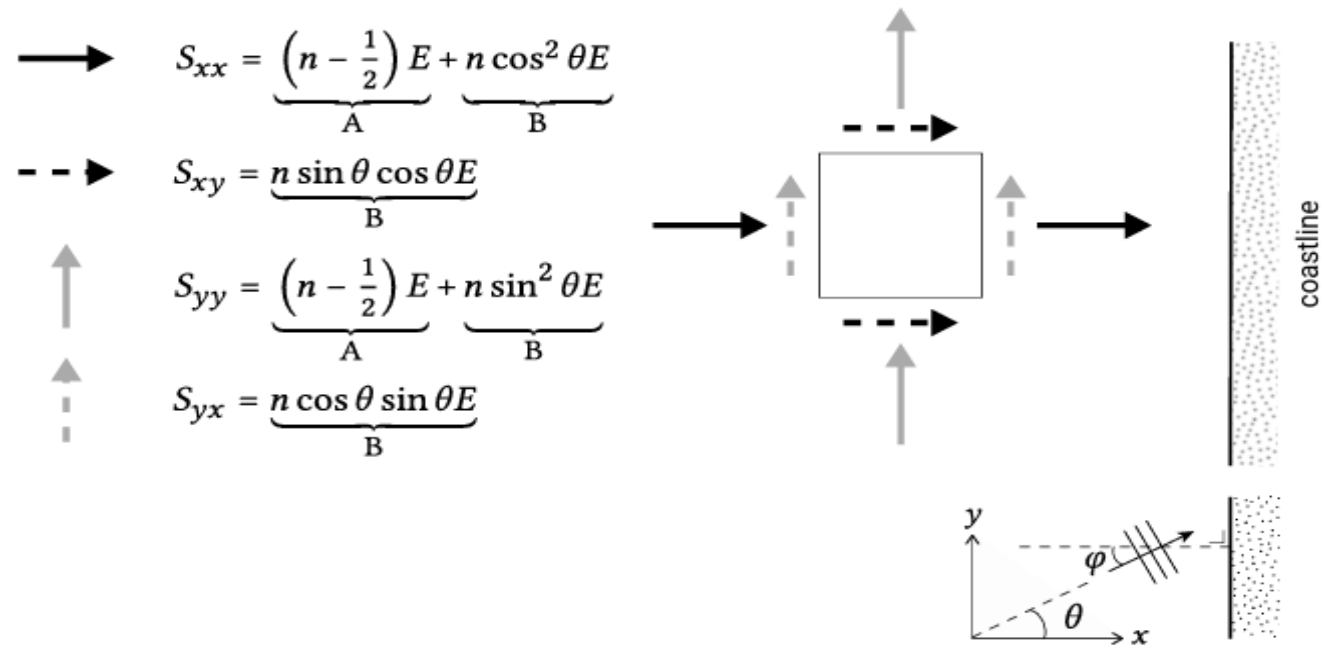
- we can now **deduce** the **more general** formulas for the **radiation stress components for waves** travelling in a **direction θ** relative to the **positive x -direction**.

- If the **x -axis** is not the **propagation direction** we have $\vec{u} = (u_x, u_y) = (u \cos \theta, u \sin \theta)$ and we find:

$$S_{xx} = \left(n - \frac{1}{2} + n \cos^2 \theta \right) E$$

$$S_{yy} = \left(n - \frac{1}{2} + n \sin^2 \theta \right) E$$

$$S_{xy} = S_{yx} = n \cos \theta \sin \theta E$$



- In the situation of an **alongshore uniform coast**, the **positive x -direction** is generally taken as the **shore-normal direction** and the **positive y -direction** is **parallel** to the **coastline**.
- The **angle of incidence** of the wave φ with respect to the **depth contours** is now **equal** to the **angle θ** with the **positive x -axis**.
- For that situation, the **radiation stress components** are sketched in **Figure**.

- If the **x -axis coincides** with the **wave propagation direction** ($\theta = 0$), the equation reduce to:

$$\begin{aligned} S_{xx} &= (2n - 1/2)E \\ S_{yy} &= (n - 1/2)E \text{ for waves propagating in the } x\text{-direction} \\ S_{xy} &= S_{yx} = 0 \end{aligned}$$

This is in accordance with the earlier findings that:

- The **normal stresses perpendicular** to the **direction of wave propagation** consist of the **pressure part** $(n - 1/2)E$ only.
- The **normal stresses in the direction of wave propagation** consist of the **pressure part** $(n - 1/2)E$ **plus** the **part due to momentum transfer** by the **horizontal orbital motion** nE .
- The **shear stresses** are **zero** if the **x -direction** is the **direction of wave propagation**.

- In **deep water**, where $n = 1/2$, the **pressure component** to the **radiation stress** is $(n - 1/2)E = 0$ and the **fluid particle part** becomes $nE = 1/2E$ in the **direction of wave propagation**.

- Thus, in the case of **wave propagation** in the **x -direction**, we find :

$$S_{xx} = 1/2E,$$

$$\text{and, } S_{yy} = 0.$$

- In **shallow water**, on the other hand, where $n = 1$, it follows that:

$$S_{xx} = 3/2E,$$

$$\text{and } S_{yy} = 1/2E.$$

- Here both contributions add to the **total radiation stress** S_{xx} and S_{yy} is **equal** to the **pressure part**.
- The **radiation (normal) stress** in **shallow water** is clearly **larger** than in **deep water**.
- Moreover, it has **two components** in **shallow water**: **parallel** and **perpendicular** to the **direction of wave propagation**.

Radiation Stress in breaking wave

- In **breaking surf zone waves**, the **effect** of the **surface roller** will be to **delay** the **momentum release** from the **wave-breaking**.
- In practice this means that the **equations** for the **radiation stresses** will contain a **term** due to **roller energy** too.
- Different models exist for this term, enhancing either the **pressure part** or **advection part** of the **radiation stress** or **both**.
- **Svendsen** (1984) proposes an **additional contribution** to the **radiation stress** in the **propagation direction** due to the **velocity** in the **roller** by an amount

$$q_{\text{roller}}^c = \alpha E_r$$

- This **contribution** is **concentrated** near the **water surface**, where the **wave-breaking** takes place.

- Under the assumption of **shallow water**, the **radiation stress** below the **wave trough level** is dominated by the **part due to advection of momentum** by the **horizontal orbital motion**.
- Since in **shallow water** the **horizontal orbital velocity** is **uniformly distributed**, the **advective part** of the **radiation stress** is **uniformly distributed** as well.

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Problem

A wave with a height $H=2$ m and a wavelength $L=50$ m approaches perpendicular to a straight coastline. The water depth h is 10m. Calculate the principal radiation stresses S_{xx} and S_{yy} . Assume the density of seawater $\rho = 1025 \text{ kg/m}^3$ and gravitational acceleration $g = 9.81 \text{ m/s}^2$. Also, explain the physical significance of these stresses in coastal processes.

Solution : Given $H = 2$ m, $L = 50$ m, $h = 10$ m, $\rho = 1025 \text{ kg/m}^3$, $g = 9.81 \text{ m/s}^2$

The principal radiation stresses S_{xx} and S_{yy} when wave approaches perpendicular to a straight coastline,

$$S_{xx} = \left(2n - \frac{1}{2}\right) E$$

$$S_{yy} = \left(n - \frac{1}{2}\right) E$$

Now ,

$$E = \frac{1}{8} \rho g H^2$$

$$E = \frac{1}{8} \times 1025 \times 9.81 \times 2^2$$

$$E = 5027.625 \text{ J/m}^2$$

$$n = \frac{1}{2} \left[1 + \frac{2kh}{\sinh 2kh} \right]$$

$$k = \frac{2\pi}{L} = \frac{2\pi}{50} = 0.126$$

$$n = \frac{1}{2} \left[1 + \frac{2 \times 0.126 \times 10}{\sinh 2 \times 0.126 \times 10} \right] = 0.704$$

$$S_{xx} = \left(2n - \frac{1}{2} \right) E$$

$$S_{xx} = \left(2 \times 0.704 - \frac{1}{2} \right) \times 5027.625$$

$$S_{xx} = 4565.08 \text{ N/m}$$

$$S_{yy} = (n - \frac{1}{2}) E$$

$$S_{yy} = (0.704 - \frac{1}{2}) \times 5027.625$$

$$S_{yy} = 1025.635 N/m$$

- It represents the excess momentum flux due to the waves and is responsible for **wave setup**, which is the rise in mean water level at the coast. In this case, $S_{xx} = 4565.08 N/m$
- It represents the lateral pressure exerted by the waves and contributes to the development of **longshore currents**. In this case, $S_{yy} = 1025.635 N/m$